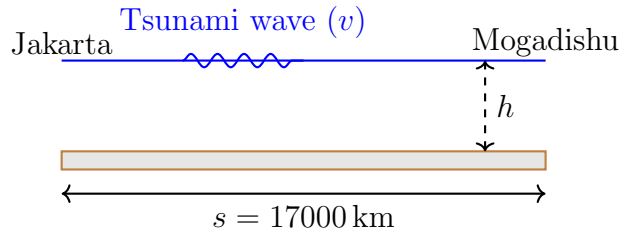


(KTXFI1EBNF) Physics I. Test 2 Solutions
May 18, 2026

1. Estimate the average depth of the Indian Ocean if a tsunami takes 24 hours to travel from Jakarta to Mogadishu. (Note: The propagation speed of shallow-water waves is $v = \sqrt{gh}$. The distance is 17000 km. $g = 9.81 \text{ m/s}^2$.)



First, we convert the given time from hours to seconds:

$$t = 24 \text{ hours} = 24 \cdot 3600 \text{ s} = 86400 \text{ s}$$

The distance s in meters is:

$$s = 17000 \text{ km} = 17 \cdot 10^6 \text{ m}$$

The average propagation speed of the tsunami wave is:

$$v = \frac{s}{t} = \frac{17 \cdot 10^6 \text{ m}}{86400 \text{ s}} \approx 196.76 \text{ m/s}$$

Using the formula for shallow-water waves:

$$v = \sqrt{gh} \implies v^2 = gh \implies h = \frac{v^2}{g}$$

Substituting the calculated velocity and $g = 9.81 \text{ m/s}^2$:

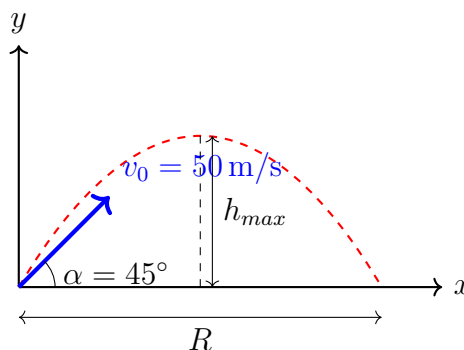
$$h = \frac{(196.76 \text{ m/s})^2}{9.81 \text{ m/s}^2} \approx \frac{38714.50 \text{ m}^2/\text{s}^2}{9.81 \text{ m/s}^2} \approx 3946.43 \text{ m}$$

The estimated average depth of the ocean along this path is approximately 3946.43 m.

(10 points)

2. A projectile is launched at an angle of 45° with an initial velocity of 50 m/s ($g = 9.81 \text{ m/s}^2$). Calculate:

- The time to reach the peak of the trajectory.
- The maximum height.
- The horizontal range (impact distance).
- The total time of flight.



The initial components of the velocity are:

$$v_{0x} = v_0 \cos(45^\circ) = 50 \text{ m/s} \cdot \frac{\sqrt{2}}{2} \approx 35.36 \text{ m/s}$$

$$v_{0y} = v_0 \sin(45^\circ) = 50 \text{ m/s} \cdot \frac{\sqrt{2}}{2} \approx 35.36 \text{ m/s}$$

(a) At the peak of the trajectory, the vertical component of the velocity becomes zero ($v_y = 0 \text{ m/s}$):

$$v_y = v_{0y} - gt_{peak} = 0 \implies t_{peak} = \frac{v_{0y}}{g} = \frac{35.36 \text{ m/s}}{9.81 \text{ m/s}^2} \approx 3.60 \text{ s}$$

(b) The maximum height can be found using the vertical motion equation:

$$h_{max} = \frac{v_{0y}^2}{2g} = \frac{(35.36 \text{ m/s})^2}{2 \cdot 9.81 \text{ m/s}^2} \approx 63.71 \text{ m}$$

(c) The horizontal range (R) is the horizontal distance traveled during the entire flight:

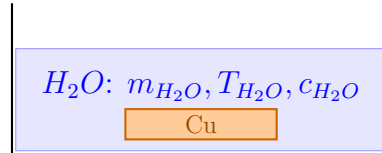
$$R = \frac{v_0^2 \sin(2\alpha)}{g} = \frac{(50 \text{ m/s})^2 \sin(90^\circ)}{9.81 \text{ m/s}^2} = \frac{2500 \text{ m}^2/\text{s}^2}{9.81 \text{ m/s}^2} \approx 254.84 \text{ m}$$

(d) Since the launch and impact points are at the same horizontal level, the total time of flight is exactly twice the time to reach the peak:

$$t_{flight} = 2 \cdot t_{peak} = 2 \cdot 3.604 \text{ s} \approx 7.21 \text{ s}$$

(10 points)

3. We place a copper piece with a mass of 100 gramms and a temperature of 40°C into 500 gramms of water at 80°C . What will be the common temperature? The specific heat capacity of copper is $3.85 \cdot 10^2 \frac{\text{J}}{\text{kg K}}$ and the specific heat capacity of water is $4.19 \cdot 10^3 \frac{\text{J}}{\text{kg K}}$.



Let us use the principle of conservation of energy:

$$m_{Cu}c_{Cu}(T_f - T_{Cu}) + m_{H_2O}c_{H_2O}(T_f - T_{H_2O}) = 0$$

Given data converted to SI base units:

$$m_{Cu} = 100 \text{ g} = 0.1 \text{ kg}, \quad T_{Cu} = 40^\circ\text{C}, \quad c_{Cu} = 385 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$m_{H_2O} = 500 \text{ g} = 0.5 \text{ kg}, \quad T_{H_2O} = 80^\circ\text{C}, \quad c_{H_2O} = 4190 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

Solving for the final temperature T_f :

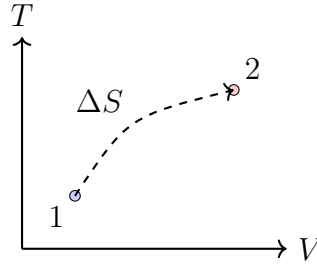
$$T_f = \frac{m_{Cu}c_{Cu}T_{Cu} + m_{H_2O}c_{H_2O}T_{H_2O}}{m_{Cu}c_{Cu} + m_{H_2O}c_{H_2O}} = \frac{0.1 \text{ kg} \cdot 385 \frac{\text{J}}{\text{kg} \cdot \text{K}} \cdot 40^\circ\text{C} + 0.5 \text{ kg} \cdot 4190 \frac{\text{J}}{\text{kg} \cdot \text{K}} \cdot 80^\circ\text{C}}{0.1 \text{ kg} \cdot 385 \frac{\text{J}}{\text{kg} \cdot \text{K}} + 0.5 \text{ kg} \cdot 4190 \frac{\text{J}}{\text{kg} \cdot \text{K}}}$$

$$T_f = \frac{1540 \frac{\text{J} \cdot ^\circ\text{C}}{\text{K}} + 167600 \frac{\text{J} \cdot ^\circ\text{C}}{\text{K}}}{38.5 \frac{\text{J}}{\text{K}} + 2095 \frac{\text{J}}{\text{K}}} = \frac{169140 \frac{\text{J} \cdot ^\circ\text{C}}{\text{K}}}{2133.5 \frac{\text{J}}{\text{K}}} \approx 79.28^\circ\text{C}$$

The common equilibrium temperature will be approximately 79.28°C .

(10 points)

4. What is the entropy change if the volume of $m = 14$ g of nitrogen changes from $V_1 = 30$ l to $V_2 = 120$ l, and its temperature changes from $T_1 = 27^\circ\text{C}$ to $T_2 = 327^\circ\text{C}$? The molar mass is $M = 28$ g/mol, $R = 8.31 \frac{\text{J}}{\text{mol}\cdot\text{K}}$.



Nitrogen (N_2) is a diatomic ideal gas. The specific heat capacity at constant volume is $C_V = \frac{5}{2}R$, where $R = 8.314 \frac{\text{J}}{\text{mol}\cdot\text{K}}$. First, calculate the number of moles (n):

$$n = \frac{m}{M} = \frac{14 \text{ g}}{28 \text{ g/mol}} = 0.5 \text{ mol}$$

Convert temperatures to Kelvin:

$$T_1 = 27^\circ\text{C} + 273.15 \text{ K} = 300.15 \text{ K}$$

$$T_2 = 327^\circ\text{C} + 273.15 \text{ K} = 600.15 \text{ K}$$

The total entropy change (ΔS) is given by:

$$\Delta S = n \left(\frac{5}{2}R \right) \ln \left(\frac{T_2}{T_1} \right) + nR \ln \left(\frac{V_2}{V_1} \right)$$

Substitute the corresponding values:

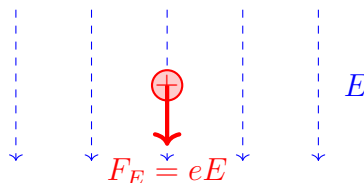
$$\Delta S = 0.5 \text{ mol} \cdot \left(2.5 \cdot 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \ln \left(\frac{600.15 \text{ K}}{300.15 \text{ K}} \right) + 0.5 \text{ mol} \cdot 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \ln \left(\frac{0.120 \text{ m}^3}{0.030 \text{ m}^3} \right)$$

$$\Delta S = 10.3925 \frac{\text{J}}{\text{K}} \cdot \ln(2) + 4.157 \frac{\text{J}}{\text{K}} \cdot \ln(4) \approx 7.204 \frac{\text{J}}{\text{K}} + 5.763 \frac{\text{J}}{\text{K}} = 12.97 \frac{\text{J}}{\text{K}}$$

The entropy change of the nitrogen gas is approximately $12.97 \frac{\text{J}}{\text{K}}$.

(10 points)

5. Determine the acceleration of a proton ($m = 1.67 \cdot 10^{-27}$ kg, $e = 1.6 \cdot 10^{-19}$ C) in an electric field of strength $E = 10000$ N/C. How many times greater is this acceleration than the acceleration due to gravity?



The electrostatic force acting on the proton is:

$$F_E = e \cdot E$$

According to Newton's second law, the acceleration a is:

$$a = \frac{e \cdot E}{m}$$

Substituting the given constants:

$$a = \frac{1.6 \cdot 10^{-19} \text{ C} \cdot 10000 \text{ N/C}}{1.67 \cdot 10^{-27} \text{ kg}} \approx 9.58 \cdot 10^{11} \text{ m/s}^2$$

To find how many times greater this is than gravity ($g = 9.81 \text{ m/s}^2$):

$$\text{Ratio} = \frac{a}{g} = \frac{9.581 \cdot 10^{11} \text{ m/s}^2}{9.81 \text{ m/s}^2} \approx 9.77 \cdot 10^{10}$$

The acceleration of the proton is approximately $9.58 \cdot 10^{11} \text{ m/s}^2$, which is roughly $9.77 \cdot 10^{10}$ times greater than gravity.

(10 points)

Requirement for obtaining the course signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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